Productivity and Wedges:

Economic Convergence and the Real Exchange Rate

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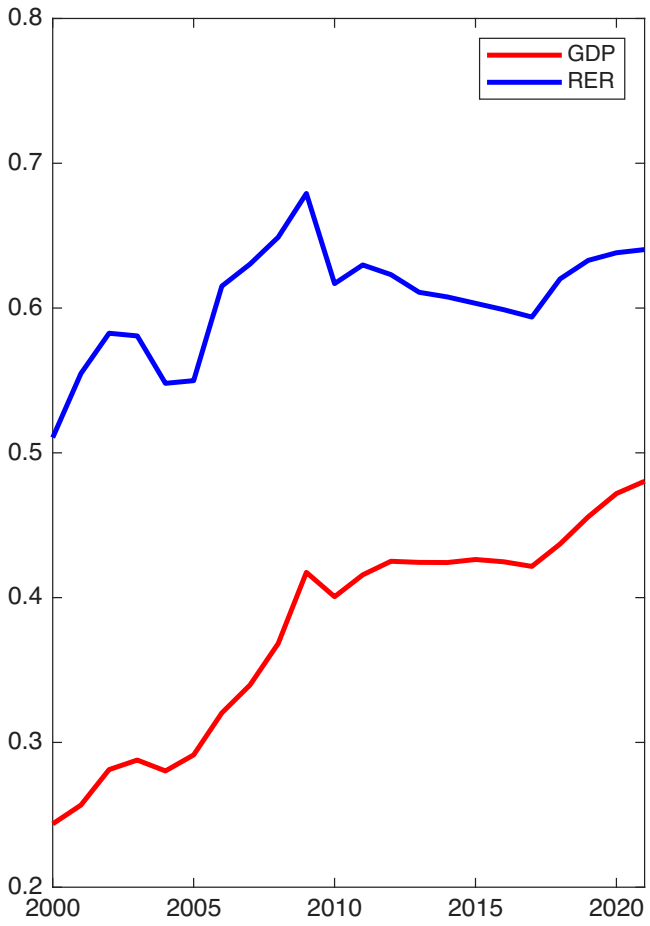
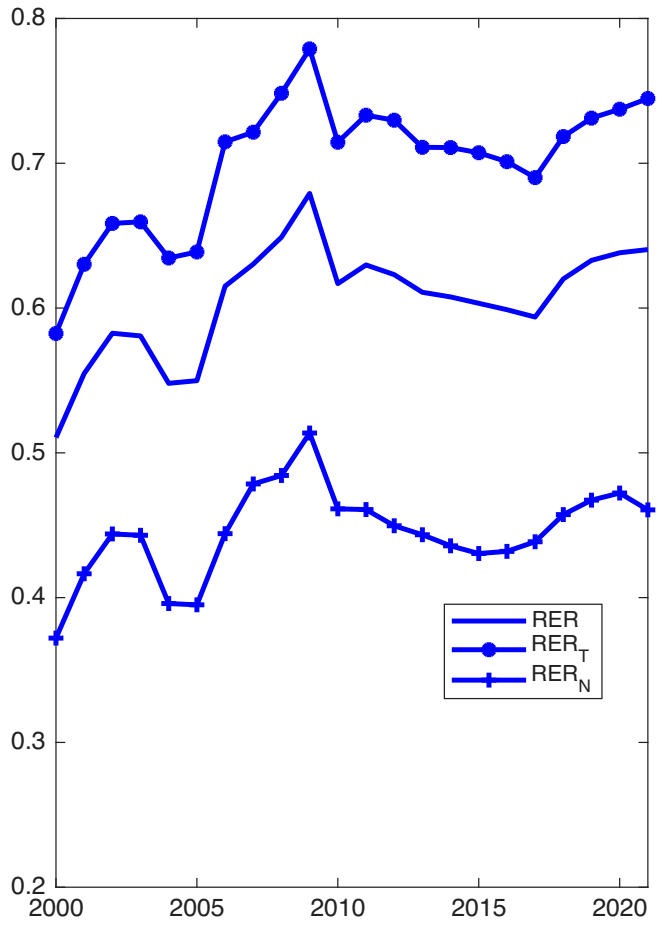
Motivation

- Wealthier countries have higher price levels — a robust global pattern known as the *Penn Effect*, but
  - Price level convergence (PPP) does not always hold among countries with similar GDP per capita
  - Significant real appreciation is observed when countries move from developing to developed status, but heterogeneity remains within income groups
- In the EU context, market integration was expected to reduce income and price level gaps across member states
  - Among Western European countries, price convergence is weak, possibly due to stable income gaps over time
  - In contrast, Eastern European countries show clear GDP convergence since joining the EU
  - But real exchange rate appreciation stalls after 2008, despite continued GDP convergence — presenting a puzzle and a policy-relevant issue for EU integration
    - This is also true for RERs of Traded and Non-Traded — another puzzle

RERs and GDP


GDP
RER
0.8
0.7
0.6
0.5
0.4
0.3
0.2
2000
2005
2010
2015
2020


RER
RER_T
RER_N
0.8
0.7
0.6
0.5
0.4
0.3
0.2
2000
2005
2010
2015
2020

What we do

- Explore the relationship between GDP convergence and real exchange rate dynamics in Eastern Europe
- Use a mix of empirical analysis and theoretical modeling to understand the divergence between income growth and price level convergence
  - Theoretically understand the dynamics of the real exchange rate with the **analytical** solution
  - Regress real exchange rate on observed sectoral productivity data and evaluate the role of
    - Sectoral productivity differentials
    - Labor market wedges
    - Capital inflows
  - Construct a multi-country, two-sector general equilibrium model to capture drivers of both GDP and real exchange rates

Key takeaways

- Theoretical
  - The association between GDP growth and the real exchange rate is nuanced (**Propositions 1** and **2**)
  - Testable implications, which are consistent with estimation results
- Empirical
  - Both productivity growth and labor market distortions jointly explain long-run real exchange rate movements
    - Empirical evidence supports a modified Balassa-Samuelson effect where traded-sector productivity appreciates and non-traded-sector productivity depreciates the real exchange rate
    - Model simulations closely match historical real exchange rate paths when accounting for productivity, labor wedges, and capital inflows
    - GDP convergence can take place without the real exchange rate appreciation
- Real exchange rate is not disconnected in the long-run

Literature review

- Balassa-Samuelson effect
  - Balassa (1964); Samuelson (1964)
- Penn effect
  - Kravis, Heston and Summers (1978)
- Empirical
  - De Gregorio, Giovanni, and Wolf (1994); Canzoneri, Cumby and Diba (1999); Choudhri and Kahn (2005); Lee and Tang (2007), Hassan (2016); Zhang (2017); Devereux and Engel (2018)
- Episodic perspective
  - Berka and Devereux (2013); Berka, Devereux and Engel (2018); Meshulam and Sanfey (2019); Ju, Lin, Liu and Shi (2020); Piton (2021); Irwin and Obstfeld (2024); Armendariz, Resende, Ferrucci, Hu, Naik and Ugur (2025); Miyamoto, Nguyen and Oh (2025)
- Long-run
  - Devereux (1999); Crucini, Yilmazkuday, and Davis (2014); Irwin and Obstfeld (2024); Cao, Evans, and Luo (2025)
- Review
  - Froot and Rogoff (1995); Rogoff (1996); Tica and Druid (2006); Morvillier (2020); Itskhoki (2021)

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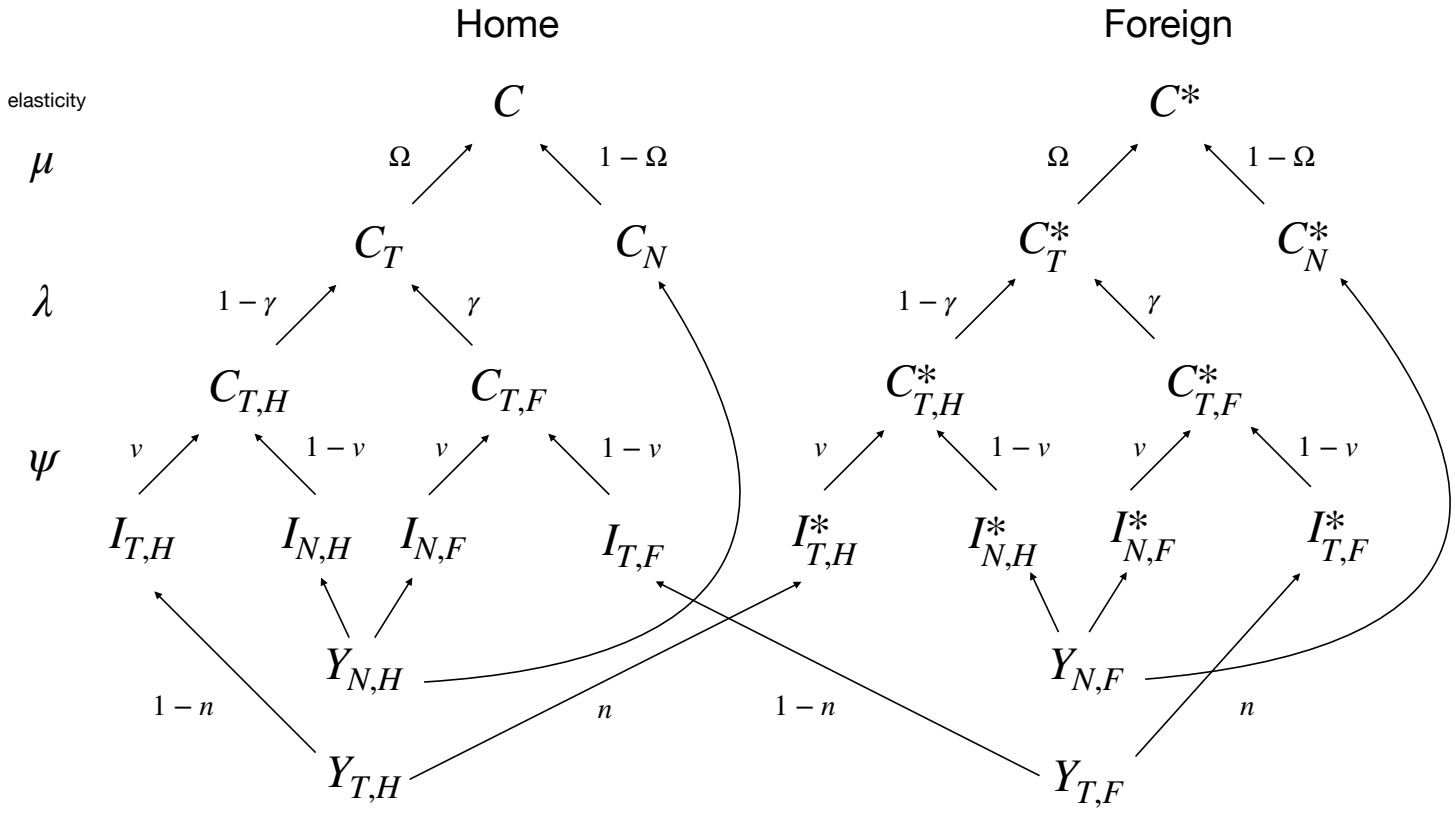
Set up

- Two countries: Home and Foreign
- Household: Consumption and labor supply
- Firms: Production and labor demand
  - Two sectors: Traded and Non-Traded
    - Production uses labor with a fixed capital stock in each sector
    - Both home and foreign traded goods require a distributional input of non-traded goods (Burstein, Eichenbaum and Rebelo, 2005, Corsetti, Dedola and Leduc, 2008, Crucini, Yilmazkuday, and Davis (2014))
- Central Bank: Nominal anchor

Supply side


Home
Foreign
elasticity
μ
λ
ψ
C
C_T
C_N
$C_{T,H}$
$C_{T,F}$
$I_{T,H}$
$I_{N,H}$
$I_{N,F}$
$I_{T,F}$
$Y_{N,H}$
$Y_{T,H}$
C^*
C^*_T
C^*_N
$C^*_{T,H}$
$C^*_{T,F}$
$I^*_{T,H}$
$I^*_{N,H}$
$I^*_{N,F}$
$I^*_{T,F}$
$Y_{N,F}$
$Y_{T,F}$
Ω
$1-\Omega$
$1-\gamma$
γ
v
$1-v$
$1-n$
n

Aggregate conditions

- Resource constraint

$$(1-n)\,Y_{T,H} = (1-n)\,I_{T,H} + nI^*_{T,H}$$

$$nY_{T,F} = (1-n)\,I_{T,F} + nI^*_{T,F}$$

$$Y_{N,H} = C_N + I_{N,H} + I_{N,F}$$

$$Y_{N,F} = C^*_N + I^*_{N,H} + I^*_{N,F}$$

- Balance of payment identity

$$PC = (1+\theta)\,P_{T,H}Y_{T,H} + P_N Y_{N,H}$$

$$B_t - \frac{B_{t+1}}{1+r_{t+1}} = \theta P_{T,H}Y_{T,H}$$

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Text book Balassa-Samuelson

- Perfect substitutability in traded goods - law of one price

$$SP_T = P_T^*$$

- Domestic labor market clearing + linear production

$$P_T = \frac{W}{A_T} \;\&\; P_N = \frac{W}{A_N} \;\Rightarrow\; \frac{P_N}{P_T} = \frac{A_T}{A_N}$$

- The real exchange rate

$$Q = \frac{SP}{P^*} = \frac{S\left[\Omega P_T^{1-\mu} + (1-\Omega) P_N^{1-\mu}\right]^{\frac{1}{1-\mu}}}{\left[\Omega \left(P_T^*\right)^{1-\mu} + (1-\Omega)\left(P_N\right)^{1-\mu}\right]^{\frac{1}{1-\mu}}} = \frac{SP_T\left[\Omega + (1-\Omega)\left(\frac{P_N}{P_T}\right)^{1-\mu}\right]^{\frac{1}{1-\mu}}}{P_T^*\left[\Omega + (1-\Omega)\left(\frac{P_N^*}{P_T^*}\right)^{1-\mu}\right]^{\frac{1}{1-\mu}}}$$

$$= \frac{\left[\Omega + (1-\Omega)\left(\frac{A_T}{A_N}\right)^{1-\mu}\right]^{\frac{1}{1-\mu}}}{\left[\Omega + (1-\Omega)\left(\frac{A_T^*}{A_N^*}\right)^{1-\mu}\right]^{\frac{1}{1-\mu}}}$$

- Log-linearized

$$\hat{Q} = (1-\Omega)\left[\left(\hat{A}_T - \hat{A}_T^*\right) - \left(\hat{A}_N - \hat{A}_N^*\right)\right] = (1-\Omega)\left(\tilde{A}_T - \tilde{A}_N\right)$$

Composite elasticity

- Home demand for home traded goods
  - Level

$$
\begin{aligned}
I_{T,H} &= v\left(\frac{P_{T,H}}{\tilde{P}_{T,H}}\right)^{-\psi} C_{T,H} = v\left(\frac{P_{T,H}}{\tilde{P}_{T,H}}\right)^{-\psi}(1-n)\left(\frac{\tilde{P}_{T,H}}{P_T}\right)^{-\lambda} C_T \\
&= v\left(\frac{P_{T,H}}{\tilde{P}_{T,H}}\right)^{-\psi}(1-n)\left(\frac{\tilde{P}_{T,H}}{P_T}\right)^{-\lambda} \Omega P_T^{-\mu} C
\end{aligned}
$$

where

$$
\tilde{P}_{T,H} = \left[v P_{T,H}^{1-\psi} + (1-v)\, P_N^{1-\psi}\right]^{\frac{1}{1-\psi}}
$$

$$
P = 1 = \left[\Omega P_T^{1-\mu} + (1-\Omega)\, P_N^{1-\mu}\right]^{\frac{1}{1-\mu}}
$$

  - Linearized

$$
\hat{I}_{T,H} = (1-n)\left[-\Gamma \hat{P}_{T,H} - (\Gamma - \Xi)\, \hat{P}_N + \hat{C}\right]
$$

where

$$
\Gamma := \psi + v\,(\lambda - \psi) = (1-v)\,\psi + v\lambda > 0
$$

$$
\Xi = v\mu + (1-v)\,(\psi - \mu) + (1-v)\,\psi
$$

Real exchange rate

- Solution of the real exchange rate

$$\begin{aligned}\hat{Q} &= \frac{\Sigma(\Gamma-1)}{\Lambda}\left[(1+\phi)\tilde{A}_T-(1-\alpha)\tilde{\Psi}_T\right] \\ &\quad -\frac{\Phi+(1-\alpha)\sigma(\Gamma-1)}{\Lambda}\left[(1+\phi)\tilde{A}_N-(1-\alpha)\tilde{\Psi}_N\right] \\ &\quad +\frac{\Sigma\Phi}{\Lambda}\hat{\theta}\end{aligned}$$

where

$$\tilde{A}_j := \hat{A}_{j,H}-\hat{A}_{j,F} \qquad \tilde{\Psi}_j := \hat{\Psi}_j-\hat{\Psi}_j^*$$

$$\Phi := (1-\alpha)+(\alpha+\phi)\Gamma > 0 \qquad \Theta := (1-\alpha)+(\alpha+\phi)\Xi > 0$$

$$\Sigma := (\sigma+\phi)+\alpha(1-\sigma) = \sigma(1-\alpha)+(\alpha+\phi) > 0$$

$$\Lambda := \Phi(\Sigma\Xi+\Theta)+(1-\alpha)(\Gamma-1)(\Sigma+\sigma\Theta)$$

- When $\Gamma > 1$, all coefficients are positive

Real exchange rate

$$\begin{aligned}\hat{Q} \quad = \quad & \frac{\Sigma\left(\Gamma-1\right)}{\Lambda}\left[\left(1+\phi\right)\tilde{A}_T-\left(1-\alpha\right)\tilde{\Psi}_T\right] \\ & -\frac{\Phi+\left(1-\alpha\right)\sigma\left(\Gamma-1\right)}{\Lambda}\left[\left(1+\phi\right)\tilde{A}_N-\left(1-\alpha\right)\tilde{\Psi}_N\right] \\ & +\frac{\Sigma\Phi}{\Lambda}\hat{\theta}\end{aligned}$$

- With higher technological progress in the traded goods sector than its foreign counterpart, the real exchange rate appreciates
- With higher technological progress in the non-traded goods sector than its foreign counterpart, the real exchange rate depreciates
- Capital inflows lead to an appreciation of the real exchange rate by increasing the domestic country's relative consumption

Why Γ matters

$$\hat{Q} = \frac{\Sigma(\Gamma - 1)}{\Lambda}\left[(1+\phi)\tilde{A}_T - (1-\alpha)\tilde{\Psi}_T\right] - \frac{\Phi + (1-\alpha)\sigma(\Gamma - 1)}{\Lambda}\left[(1+\phi)\tilde{A}_N - (1-\alpha)\tilde{\Psi}_N\right] + \frac{\Sigma\Phi}{\Lambda}\hat{\theta}$$

$$\Gamma := \psi + v\,(\lambda - \psi) = (1-v)\,\psi + v\lambda > 0$$

- Higher $\tilde{A}_T$ raises income ⇒ higher demand for Non-Traded ⇒ higher P_N at Home
  - **Perfect substitutability** ($\Gamma \to \infty$): ToT fixed ⇒ appreciation - classic Balassa–Samuelson effect
    - Uniform productivity increase: $\tilde{A}_T = \tilde{A}_T = \tilde{A}$

$$\lim_{\lambda\to\infty} \hat{Q}_t = -\lim_{\lambda\to\infty} \frac{(1+\phi)^2}{\Lambda}\tilde{A} = 0$$

  - **Imperfect substitutability** ($\lambda < \infty$) and non-traded inputs in Traded ($1 - v > 0$, $\psi < \infty$): ToT deteriorates and *dampens* appreciation
  - **Knife-edge** ($\Gamma = 1$, Cobb–Douglas nests): ToT fully adjusts to share gains ⇒ Balassa–Samuelson effect *disappears* (Cole and Obstfeld, 1991)
  - **Complementarity** ($\Gamma < 1$): low trade & T & NT elasticity ⇒ Balassa–Samuelson effect can be *reversed*

Real exchange rate

$$
\begin{aligned}
\hat{Q} \quad = \quad & \frac{\Sigma\left(\Gamma-1\right)}{\Lambda}\left[\left(1+\phi\right)\tilde{A}_T-\left(1-\alpha\right)\tilde{\Psi}_T\right] \\
& -\frac{\Phi+\left(1-\alpha\right)\sigma\left(\Gamma-1\right)}{\Lambda}\left[\left(1+\phi\right)\tilde{A}_N-\left(1-\alpha\right)\tilde{\Psi}_N\right] \\
& +\frac{\Sigma\Phi}{\Lambda}\hat{\theta}
\end{aligned}
$$

- With higher technological progress in the traded goods sector than its foreign counterpart, the real exchange rate appreciates
- With higher technological progress in the non-traded goods sector than its foreign counterpart, the real exchange rate depreciates
- Capital inflows lead to an appreciation of the real exchange rate by increasing the domestic country's relative consumption

Real exchange rate

$$\begin{aligned}\hat{Q} \quad = \quad & \frac{\Sigma\left(\Gamma-1\right)}{\Lambda}\left[\left(1+\phi\right)\tilde{A}_T-\left(1-\alpha\right)\tilde{\Psi}_T\right] \\ & -\frac{\Phi+\left(1-\alpha\right)\sigma\left(\Gamma-1\right)}{\Lambda}\left[\left(1+\phi\right)\tilde{A}_N-\left(1-\alpha\right)\tilde{\Psi}_N\right] \\ & +\frac{\Sigma\Phi}{\Lambda}\hat{\theta}\end{aligned}$$

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Proposition 1: Balassa-Samuelson Hypothesis

Proposition

Balassa-Samuelson Hypothesis

(1) A positive productivity shock in the traded goods sector generates real exchange rate appreciation (no change) (depreciation) when $\Gamma > 1$, $(\Gamma = 1)$, $(\Gamma < 1)$

(2) A uniform increase in productivity growth across sectors leads to a real exchange rate depreciation when $\Gamma < \infty$

(3) Let $\tilde{A}_T = \tilde{A}_N + \delta \equiv \tilde{A} + \delta$, *where* $\delta > 0$. *Higher productivity in traded goods than in non-traded goods is associated with a real exchange rate appreciation if and only if*

$$\Sigma\left(\Gamma - 1\right)\delta - \left(1 + \phi\right)\tilde{A} > 0$$

- The impact of the traded goods productivity growth exceeds the negative effect of the uniform growth rate

Proposition 2: The Penn Effect

Proposition

The Penn Effect

Controlling for productivity shocks and the labor wedge in the non-traded sector, there is a positive correlation between relative GDP per capita and the real exchange rate

- Relative GDP and the real exchange rate will move in the same direction

$$\tilde{Q} = \frac{\Sigma\left(\hat{C} - \hat{C}^*\right) - \left[(1+\phi)\,\tilde{A}_N - (1-\alpha)\,\tilde{\Psi}_N\right]}{\Theta}$$

  - $\frac{\Sigma}{\Theta}$ is independent of Γ: even if the Balassa-Samuelson effect fails, the Penn effect may still be observed
  - Backus-Smith puzzle
    - Complete market: negative correlation btw $\tilde{Q}$ and $\left(\hat{C} - \hat{C}^*\right)$
    - Data: positive correlation btw $\tilde{Q}$ and $\left(\hat{C} - \hat{C}^*\right)$, implying incomplete risk sharing

Nuanced effects

- **Propositions 1** and **2** imply that the association between GDP growth and the real exchange rate is nuanced
  - Even when GDP is driven by productivity growth in the traded goods sector, it may not be combined with real exchange rate appreciation
  - This same lack of coincidence between output growth and real exchange rate appreciation also holds at the sectoral level
    - Implications for convergence in GDP and RER

RERs of Traded and Non-Traded

- The real exchange rate of Traded

$$\hat{Q}_T = \hat{S} + \left(\hat{P}_T - \hat{P}_T^*\right) = 2\left(1 - v\right)\hat{Q}$$

- The real exchange rate of Non-Traded

$$\hat{Q}_N = \hat{S} + \left(\hat{P}_N - \hat{P}_N^*\right) = 2\hat{Q}$$

- Consistent with East European countries' experiences
  - Co-movement among real exchange rate of aggregate, Traded and Non-Traded
- The text book Balassa-Samuelson effect (when $v = 1$) leads to

$$\hat{Q}_T = 0$$

Edgeworth complements / substitutes

- Solution of the relative output of Non-Traded

$$\begin{aligned}\hat{y}_N &= -\frac{(1-\alpha)(\Gamma-1)(\sigma\Xi-1)}{\Lambda}\left[(1+\phi)\tilde{A}_T-(1-\alpha)\tilde{\Psi}_T\right] \\ &\quad +\frac{(1-\alpha)(\Gamma-1)(\sigma\Xi+1)+2\Phi\Xi}{\Lambda}\left[(1+\phi)\tilde{A}_N-(1-\alpha)\tilde{\Psi}_N\right] \\ &\quad -\frac{(1-\alpha)\Phi(\sigma\Xi-1)}{\Lambda}\hat{\theta}\end{aligned}$$

- Solution of the relative output of Traded

$$\begin{aligned}\hat{y}_T &= \frac{\Gamma\left[(1-\alpha)+\Sigma\Xi\right]}{\Lambda}\left[(1+\phi)\tilde{A}_T-(1-\alpha)\tilde{\Psi}_T\right] \\ &\quad -\frac{\Gamma(1-\alpha)(\sigma\Xi-1)}{\Lambda}\left[(1+\phi)\tilde{A}_N-(1-\alpha)\tilde{\Psi}_N\right] \\ &\quad -\frac{\Gamma(1-\alpha)(\Sigma+\sigma\Theta)}{\Lambda}\hat{\theta}\end{aligned}$$

- $\sigma\Xi < (>) 1$ implies *Edgeworth complements* (*substitutes*)

$$\Xi = v\mu + (1-v)(\psi-\mu) + (1-v)\psi$$

  - Productivity growth in T and NT moves
    - the real exchange to opposite directions
    - the sectoral outputs to the same direction (opposite directions)

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Data generating process

- Suppose that the data-generating process for the real exchange rate is given by the solution

$$\hat{Q} = \bar{A}\tilde{A}_T - \bar{B}\tilde{A}_N - \bar{C}\tilde{\Psi}_T + \bar{D}\tilde{\Psi}_N + \bar{E}\hat{\theta},$$

where

$$\bar{A} := \frac{\Sigma\left(\Gamma - 1\right)\left(1 + \phi\right)}{\Lambda} > 0 \qquad \bar{B} := \frac{\left[\Phi + \left(1 - \alpha\right)\sigma\left(\Gamma - 1\right)\right]\left(1 + \phi\right)}{\Lambda} > 0$$

$$\bar{C} := \frac{\Sigma\left(\Gamma - 1\right)\left(1 - \alpha\right)}{\Lambda} > 0 \qquad \bar{D} := \frac{\left[\Phi + \left(1 - \alpha\right)\sigma\left(\Gamma - 1\right)\right]\left(1 - \alpha\right)}{\Lambda} > 0$$

$$\bar{E} := \frac{\Sigma\Phi}{\Lambda} > 0$$

- If we have data counterparts for all exogenous variables, we can recover the coefficients $\bar{A}$, $\bar{B}$, $\bar{C}$, $\bar{D}$ and $\bar{E}$ via OLS

Omitted variables

- Data counterparts for labor wedges $\tilde{\Psi}_T$ and $\tilde{\Psi}_N$ are not available
- Estimate the real exchange rate equation omitting these variables

$$\hat{Q} = \hat{a}\tilde{A}_T + \hat{b}\tilde{A}_N + \hat{e}\hat{\theta} + \varepsilon$$

- We can still recover the true coefficients $\bar{A}$, $\bar{B}$, but the variances of the estimates are

$$\mathrm{Var}(\hat{a}) = \mathrm{Var}(\hat{b}) = \mathrm{Var}(\hat{e}) = \frac{\bar{C}^2 + \bar{D}^2}{n}$$

ULC as a proxy for labor wedge

- The solution for the aggregate ULC is given by

$$\hat{u} = \bar{F}\tilde{A}_T - \bar{G}\tilde{A}_N - \bar{H}\tilde{\Psi}_T + \bar{I}\tilde{\Psi}_N + \bar{J}\hat{\theta}$$

where

$$\bar{F} := \left[-\frac{\Sigma\Xi + \Theta}{2\Lambda} + \frac{\Sigma\left(\Gamma - 1\right)}{\Lambda}\right]\left(1 + \phi\right)$$

$$-\bar{G} := \left[\frac{\left(1-\alpha\right)\left(\sigma\Xi - 1\right)}{2\Lambda} - \frac{\left(1-\alpha\right)\left(1-\sigma\right) + \Sigma\Gamma}{\Lambda}\right]\left(1 + \phi\right)$$

$$-\bar{H} := \left[\frac{\Sigma\Xi + \Theta}{2\Lambda} - \frac{\Sigma\left(\Gamma - 1\right)}{\Lambda}\right]\left(1 - \alpha\right)$$

$$\bar{I} := \left[-\frac{\left(1-\alpha\right)\left(\sigma\Xi - 1\right)}{2\Lambda} + \frac{\left(1-\alpha\right)\left(1-\sigma\right) + \Sigma\Gamma}{\Lambda}\right]\left(1 - \alpha\right)$$

$$\bar{J} := \left[\frac{\left(1-\alpha\right)\left(\Sigma + \sigma\Theta\right)}{2\Lambda} + \frac{\Sigma\Phi}{\Lambda}\right] > 0$$

- All coefficients are positive when the trade elasticity is relatively high: $\Gamma = \left(1 - v\right)\psi + v\lambda$

ULC as a proxy for labor wedge

- Estimate the real exchange rate on $\hat{u}$ in addition to $\tilde{A}_T$, $\tilde{A}_N$ and $\hat{\theta}$

$$\hat{Q} = \tilde{a}\tilde{A}_T + \tilde{b}\tilde{A}_N + \tilde{c}\hat{u} + \tilde{e}\hat{\theta} + \varepsilon$$

- Estimated parameters are given by

$$\begin{aligned}
\tilde{a} &= \bar{A} - \tilde{c}\bar{F} \\
\tilde{b} &= -\bar{B} + \tilde{c}\bar{G} \\
\tilde{c} &= \frac{\bar{C}\bar{H} + \bar{D}\bar{I}}{\bar{H}^2 + \bar{I}^2} > 0 \\
\tilde{d} &= \bar{E} - \tilde{c}\bar{J}
\end{aligned}$$

Testable implications

$$\hat{Q} = \hat{a}\tilde{A}_T + \hat{b}\tilde{A}_N + \hat{e}\hat{\theta} + \varepsilon$$
$$\hat{Q} = \tilde{a}\tilde{A}_T + \tilde{b}\tilde{A}_N + \tilde{c}\hat{u} + \tilde{e}\hat{\theta} + \varepsilon$$

1. The sign of coefficient is positive (negative) on the traded (non-traded) technology
$$\hat{a}, \tilde{a} > 0, \quad \hat{b}, \tilde{b} < 0$$
2. Coefficients on non-traded technology are larger in absolute value than those on traded technology
$$|\hat{a}| < |\hat{b}|, \quad |\tilde{a}| < |\tilde{b}|$$
3. Coefficients exhibit consistent signs across RER, RERT and RERN
4. The absolute value of coefficients on relative technologies is largest when the dependent variable is RERN
5. Coefficients are smaller in absolute value when ULC is used as an additional explanatory variable
$$|\hat{a}| > |\tilde{a}|, \quad |\hat{b}| > |\tilde{b}|$$

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Data overview

- Annual panel data from 1999–2020 for 12 Eastern European countries
- Countries: Bulgaria, Cyprus, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Romania, Slovenia, Slovakia
- Focus on real exchange rates and labor market fundamentals
- Comparison benchmark: EU27 and in some cases EU15 countries

Sectoral classification and tradability

NACE Rev. 2 Sector Letter	Sector Name	Tradability
A	Agriculture, forestry and fishing	T
B	Mining and quarrying	T
C	Manufacturing	T
D	Electricity, gas, steam and air conditioning supply	N
E	Water supply; sewerage, waste management and remediation activities	N
F	Construction	N
G	Wholesale and retail trade	T
H	Transportation and storage	N
I	Accommodation and food service activities	N
J	Information and communication	N
K	Financial and insurance activities	N
M	Professional, scientific and technical activities	N
N	Administrative and support service activities	N
R	Arts, entertainment and recreation	N
S	Other service activities	N
B-E	Industry (except construction)	T
G-I	Wholesale and retail trade, transport, accommodation and food services	N
M-N	Professional, scientific. services	N
R-U	Arts, entertainment, recreation, and other services	N

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Regression specification

- Estimated equations

$$\log \mathrm{RER}^j_{it} = \alpha_0 + \beta_1 \log \underbrace{\left(\frac{\frac{realLabProd_{T,it}}{realLabProd_{N,it}}}{\frac{realLabProd_{T,EUt}}{realLabProd_{N,EUt}}} \right)}_{RLprod} + \beta_2 \log \underbrace{\left(\frac{RealULC_{it}}{RealULC_{EUt}} \right)}_{RULC} + \epsilon_{ti}$$

$$\log \mathrm{RER}^j_{it} = \alpha_0 + \beta_1 \log Lprod_{T,it} + \beta_2 \log Lprod_{N,it} + \beta_3 \log RULC_{it} + \epsilon_{ti}$$

Real exchange rate: Baseline

	Fixed effects				Random effects			
	(1a)	(1b)	(1c)	(1d)	(2a)	(2b)	(2c)	(2d)
RLprod	**0.189*****	**0.192*****	—	—	0.068	0.08*	—	—
	(0.05)	(0.05)			(0.05)	(0.05)		
LprodT	—	—	**0.186*****	**0.19*****	—	—	**0.095***	**0.107***
			(0.07)	(0.06)			(0.06)	(0.06)
LprodN	—	—	−**0.193*****	−**0.195*****	—	—	−0.05	−0.067
			(0.07)	(0.07)			(0.07)	(0.07)
RULC	**0.283*****	**0.284*****	**0.286*****	**0.286*****	**0.246*****	**0.25*****	**0.232*****	**0.238*****
	(0.04)	(0.03)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)	(0.04)
LR	0.0016	0.0039	0.0016	0.0039	0.0019	0.0039	0.0019	0.0040
	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)	(0.003)
$LR \times GFC$	—	−0.0048	—	−0.0048	—	−0.0042	—	−0.0043
		(0.003)		(0.003)		(0.003)		(0.003)
R^2 *(adj.)*	0.63	0.64	0.63	0.64	0.9	0.9	0.9	0.89
N	214	214	214	214	214	214	214	214
HT					reject	reject	reject	reject

Real exchange rate for Traded: Baseline

	Fixed effects				Random effects			
	(1a)	(1b)	(1c)	(1d)	(2a)	(2b)	(2c)	(2d)
RLprod	**0.136*****	**0.137*****	—	—	0.042	0.052	—	—
	(0.05)	(0.05)			(0.04)	(0.04)		
LprodT	—	—	**0.158*****	**0.159*****	—	—	**0.084***	**0.095***
			(0.06)	(0.06)			(0.05)	(0.05)
LprodN	—	—	−0.1	−0.101	—	—	0.004	−0.009
			(0.07)	(0.07)			(0.06)	(0.06)
RULC	**0.267*****	**0.268*****	**0.242*****	**0.242*****	**0.233*****	**0.237*****	**0.202*****	**0.207*****
	(0.03)	(0.03)	(0.05)	(0.05)	(0.03)	(0.03)	(0.04)	(0.04)
LR	**0.0030***	0.0040	**0.0030***	0.0041	**0.0033***	0.0040	**0.0032***	0.0041
	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)	(0.003)
$LR \times GFC$	—	−0.0020	—	−0.0021	—	−0.0015	—	−0.0017
		(0.003)		(0.003)		(0.003)		(0.003)
R^2 *(adj.)*	0.64	0.64	0.64	0.64	0.91	0.9	0.91	0.9
N	214	214	214	214	214	214	214	214
HT					reject	reject	reject	reject

Real exchange rate for Non-Traded: Baseline

	Fixed effects				Random effects			
	(1a)	(1b)	(1c)	(1d)	(2a)	(2b)	(2c)	(2d)
RLprod	**0.269****	**0.277*****	—	—	0.124	**0.138***	—	—
	(0.11)	(0.1)			(0.08)	(0.08)		
LprodT	—	—	0.197	**0.207***	—	—	0.096	0.107
			(0.12)	(0.12)			(0.09)	(0.09)
LprodN	—	—	−**0.387*****	−**0.393*****	—	—	−0.135	−0.152
			(0.14)	(0.14)			(0.11)	(0.11)
RULC	**0.356*****	**0.359*****	**0.441*****	**0.442*****	**0.332*****	**0.335*****	**0.344*****	**0.349*****
	(0.07)	(0.07)	(0.1)	(0.09)	(0.06)	(0.06)	(0.07)	(0.07)
LR	−0.0010	0.0048	−0.0010	0.0046	−0.0004	0.0051	−0.0003	0.0051
	(0.003)	(0.005)	(0.003)	(0.005)	(0.003)	(0.005)	(0.003)	(0.005)
LR × *GFC*	—	−**0.0117****	—	−**0.0114****	—	−**0.0112****	—	−**0.0111****
		(0.005)		(0.005)		(0.005)		(0.005)
R^2 (*adj.*)	0.47	0.48	0.48	0.49	0.78	0.78	0.79	0.79
N	214	214	214	214	214	214	214	214
HT					reject	reject	reject	reject

Consistency with testable implications

		Panel specifications				
		Simplest	Baseline	Fiscal	C. infl.	R. GDP
1	$\beta_T > 0$, $\beta_N < 0$	✓	✓	✓	✓	✓
2	$\lvert\beta_N\rvert > \lvert\beta_T\rvert$	×	✓−	✓−	×	✓
3	Consistent coef. signs	✓	✓	✓	✓	✓
4	Larger coefs. with RER_N;	×	✓	✓	✓	✓
5	Coefs. smaller in abs. with ULC	✓	n.a.	n.a.	n.a.	n.a.

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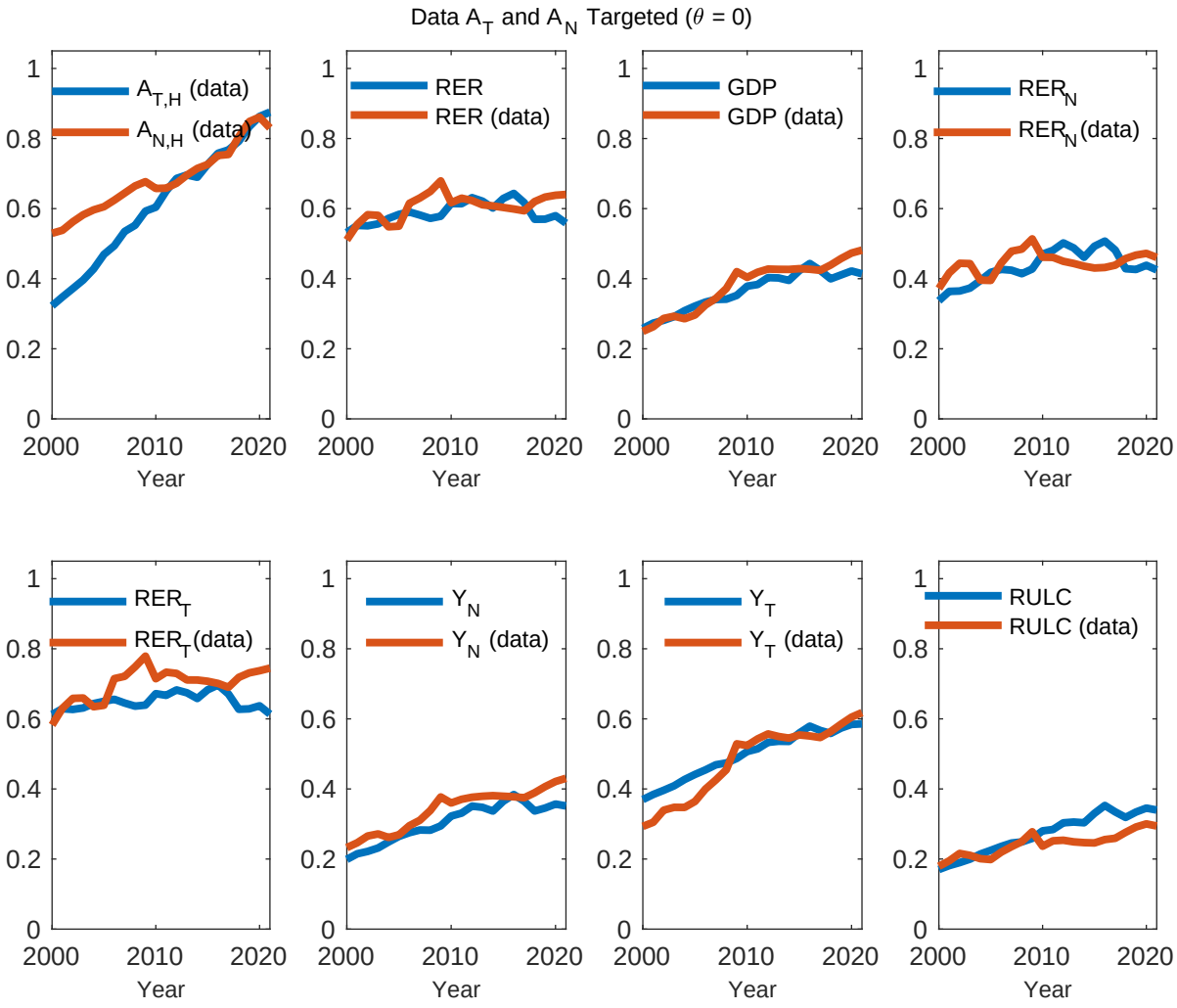
Quantitative evaluation

- Evaluate how well the model explains the historical trends in real exchange rate, GDP, and other macro variables in Eastern Europe
- Deliberately minimalist
  - Two countries, two sectors, no frictions, and only labor wedges as distortions
  - Homothetic preference: income elasticities do not influence outcomes
- Uses observed sectoral labor productivity as input data for simulations
- Model performance is evaluated by how closely it replicates both levels and changes in the observed data across the full sample period

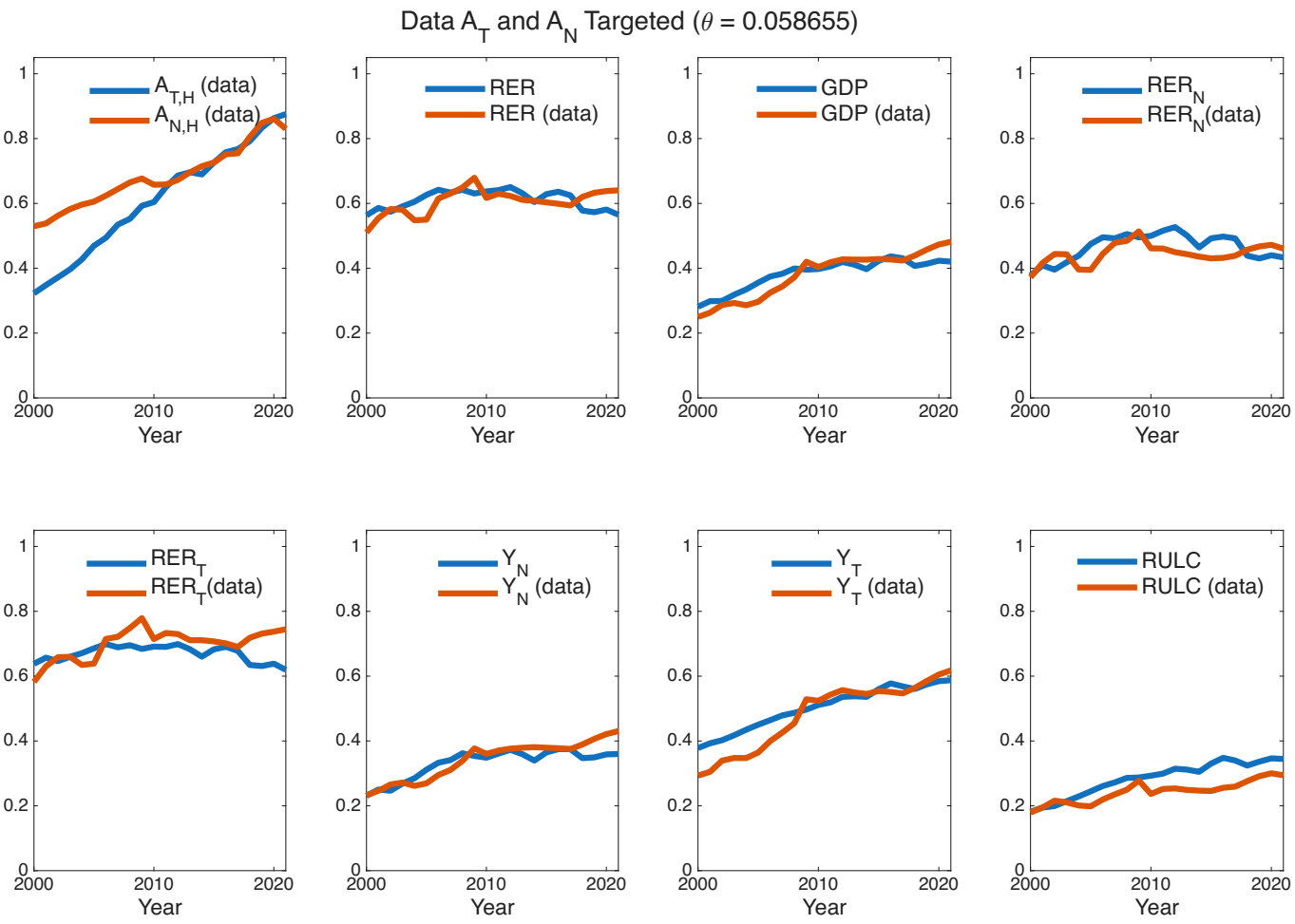
Calibration

	Parameter
$n = 0.78$	Size of Western European countries
$\Omega = 0.75$	Weight on traded goods
$v = 0.5$	Share of wholesale traded good in C_T
$\lambda = 8$	Elasticity of substitution between H and F retail traded goods
$\mu = 0.25$	Elasticity of substitution between traded good and retail service
$\psi = 0.7$	Elasticity of substitution between traded and non-traded good
$\sigma = 2$	Inverse elasticity of inter-temporal substitution
$\phi = 1$	Inverse Frisch elasticity of labor supply
$\alpha = \frac{1}{3}$	Labor share (1 minus)
$\theta = \{0, 0.059\}$	Capital flows

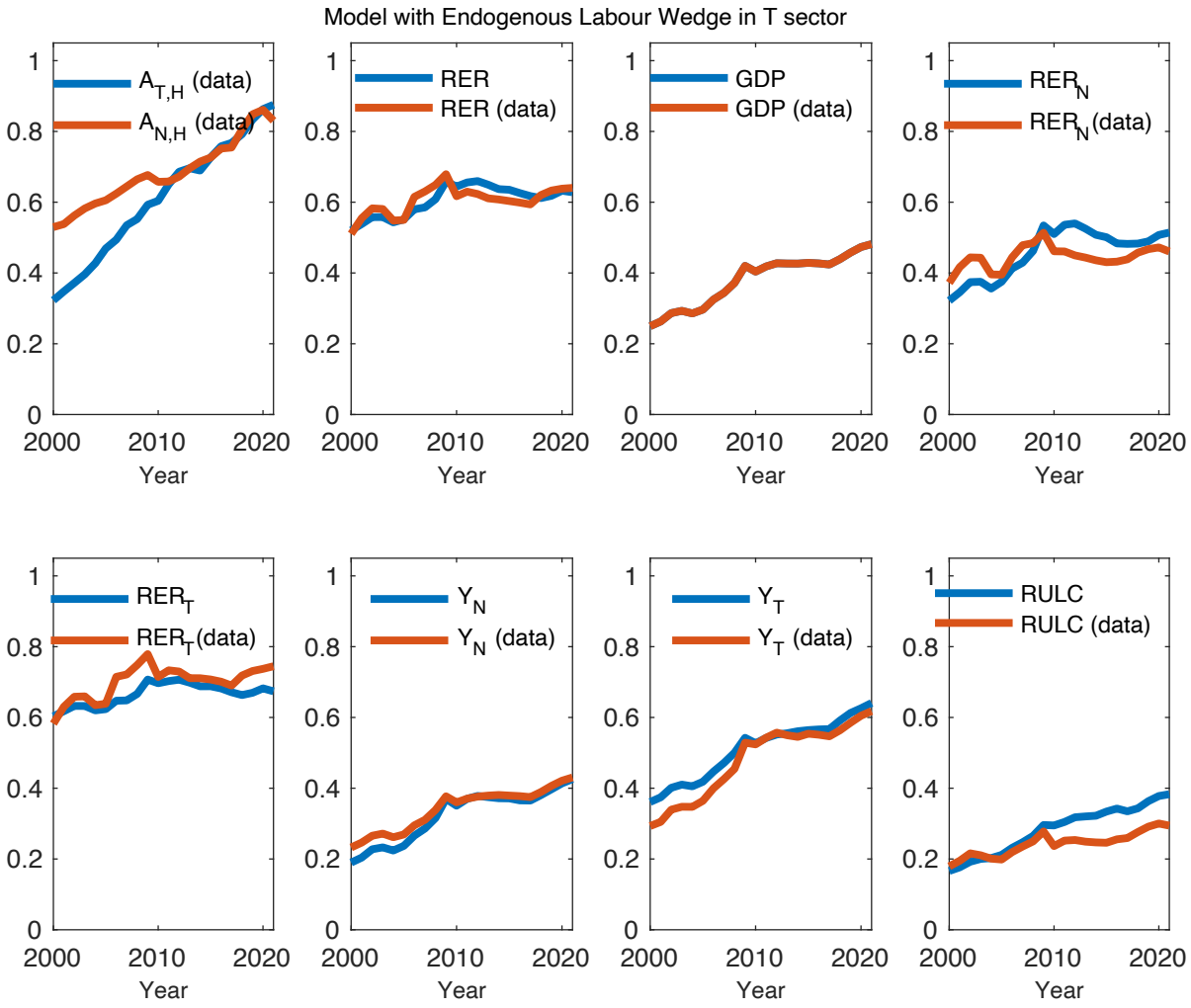
Productivities



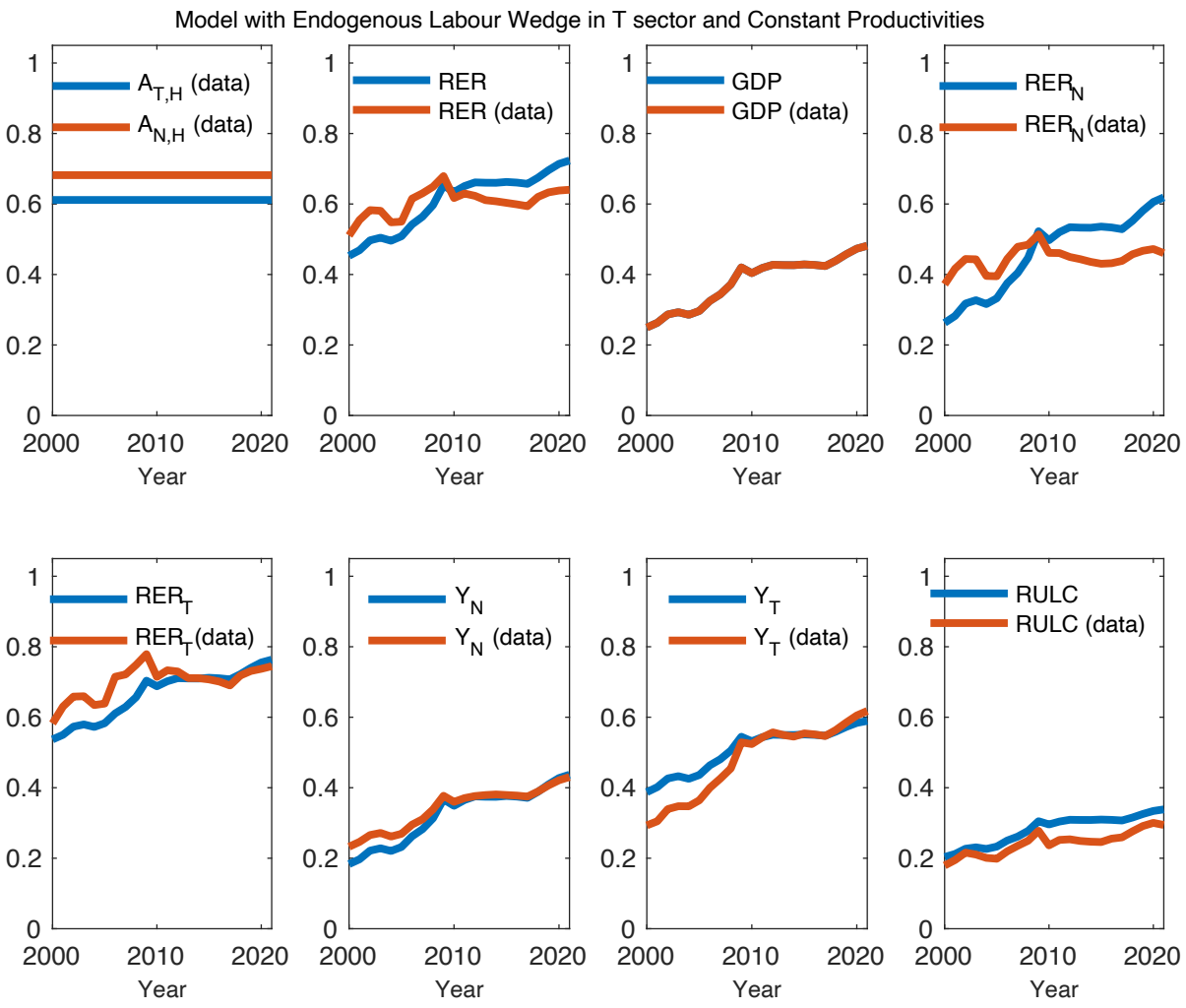
Productivities + Capital flow



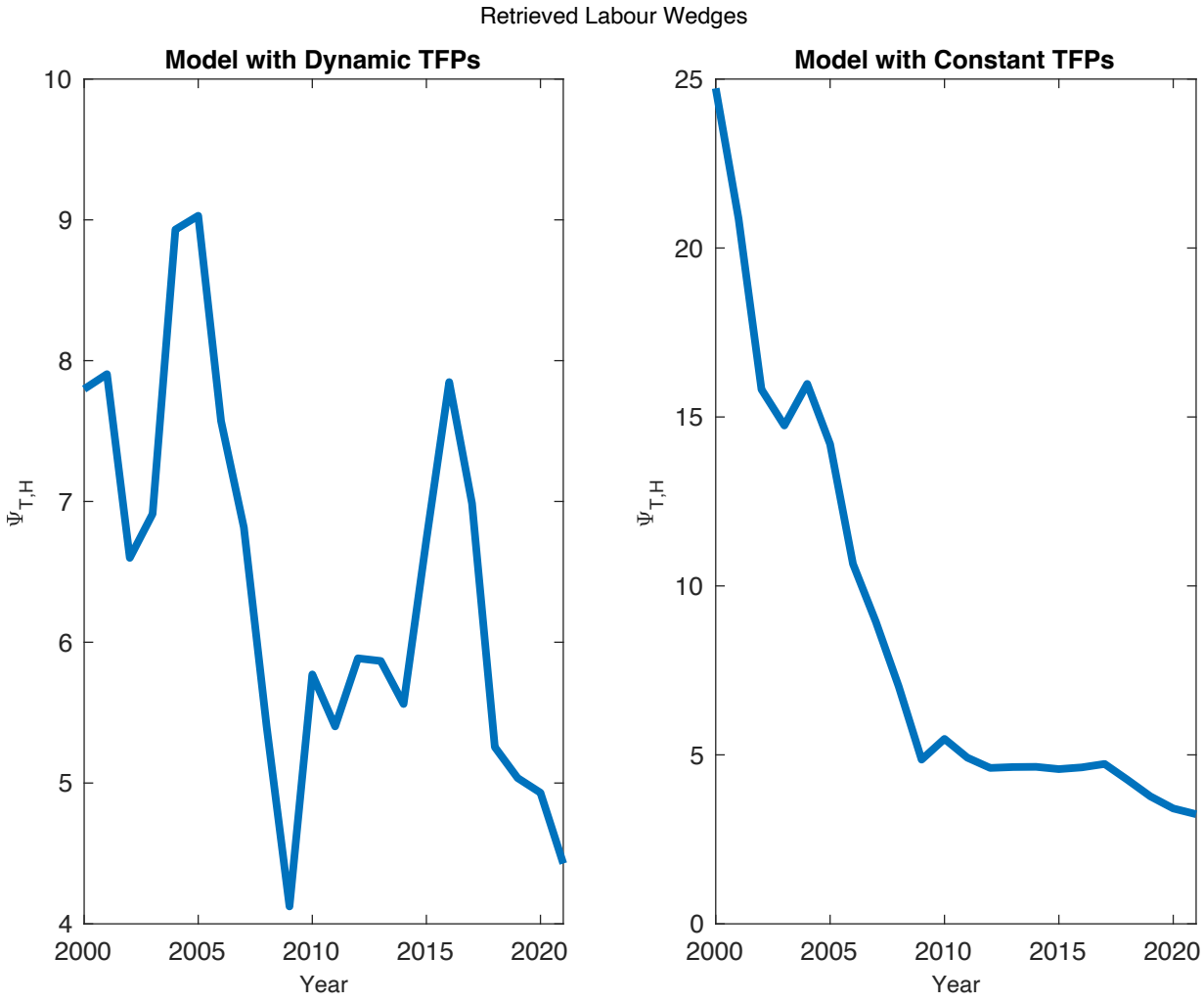
Productivities + End. labor wedge in Traded



End. labor wedge in Traded



Retrieved labor wedges in Traded



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Key takeaways

- Theoretical
  - The association between GDP growth and the real exchange rate is nuanced (**Propositions 1** and **2**)
  - Testable implications, which are consistent with estimation results
- Empirical
  - Both productivity growth and labor market distortions jointly explain long-run real exchange rate movements
    - Empirical evidence supports a modified Balassa-Samuelson effect where traded-sector productivity appreciates and non-traded-sector productivity depreciates the real exchange rate
    - Model simulations closely match historical real exchange rate paths when accounting for productivity, labor wedges, and capital inflows
    - GDP convergence can take place without the real exchange rate appreciation
- Real exchange rate is not disconnected in the long-run